

Technical Implementation Plan for Renewal & Upsell Advisor

Autonomous Revenue Agent Strategy

February 11, 2026

Goal

To build an Autonomous Revenue Agent that proactively manages renewals and upsells through a 24-hour analysis loop, Yagna-style triggers, and closed-loop voice outreach.

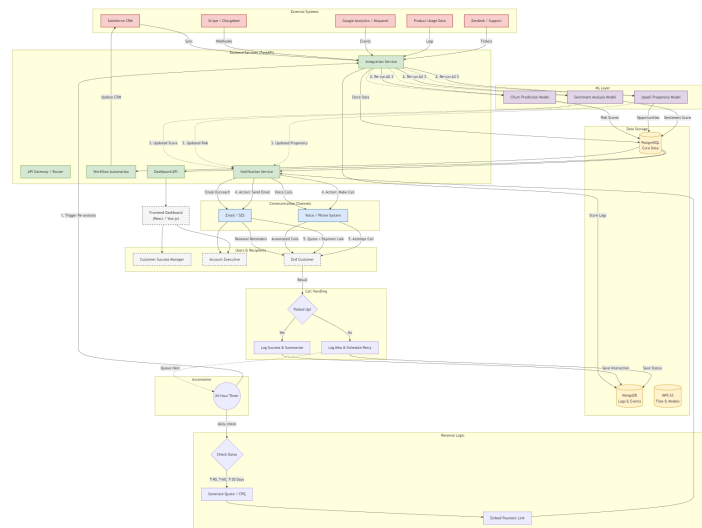


Figure 1: High-Level System Architecture

1 Phase 1: Foundation & Data Infrastructure

1.1 Database Schema (PostgreSQL)

- **Accounts Table:** `account_id`, `arr`, `contract_end_date`, `health_score`.
- **Interactions Table:** `id`, `type` (EMAIL/CALL), `status` (SENT/PICKED_UP/MISSED), `transcript`.
- **Opportunities Table:** Tracks automated renewal and upsell deals.

1.2 Event Logging (MongoDB)

- Set up a `logs` collection for unstructured data (voice call transcripts, raw email bodies, webhook payloads).

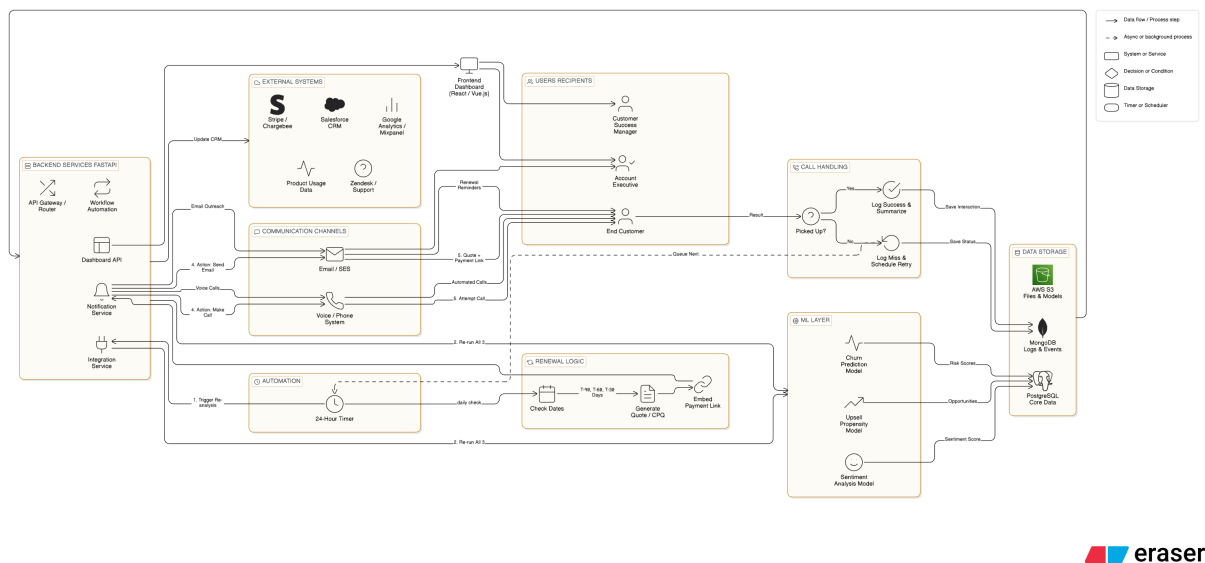
1.3 Integration Service (FastAPI)

- Implement **Salesforce Sync** (Bidirectional logic for Accounts/Contacts).
- Implement **Stripe Webhook Listener** to capture payment success/failure events in real-time.

2 Phase 2: The 24-Hour Analysis Engine

2.1 Scheduler Setup

- Configure a daily job (using APScheduler or Celery Beat) to trigger at 00:00 UTC.



2.2 AI Models Implementation

1. Sentiment Analysis (NLP):

- Integrate HuggingFace `distilbert-base-uncased-finetuned-sst-2-english`.
- **Task:** Score all emails/tickets from the last 24 hours.

2. Churn Prediction (XGBoost):

- **Inputs:** Usage drops (login frequency) + recent Sentiment Scores.
- **Output:** Risk Probability (0-100%).

3. Upsell Propensity (Collaborative Filtering):

- **Logic:** Identify accounts with > 85% license utilization.
- **Action:** Flag for "Seat Expansion" campaign.

3 Phase 3: Yagna-Style Notification Triggers

3.1 Automated Workflow Logic

The Scheduler will check contract dates daily and trigger specific actions:

T-90 Days (Early Warning): Generate preliminary Quote PDF. Send "Upcoming Renewal" email.

T-60 Days (Adoption Check): Check license usage. If usage < 50%, trigger "Adoption Intervention" playbook instead of a quote.

T-30 Days (Urgent Action): Generate final Quote. **Embed Stripe Payment Link** directly in the email for one-click renewal.

4 Phase 4: Closed-Loop Voice Agent

4.1 Voice System (Twilio / Amazon Connect)

- Implement programmable voice calls for high-value accounts.
- **Script Logic:** "Hello [Name], your renewal is due on [Date]. Do you have 2 minutes to discuss?"

4.2 Outcome Handling Logic

1. Picked Up:

- Record conversation transcript.
- Save to MongoDB `interactions` collection.
- Tag status as `SUCCESS`.

2. Missed / No Answer:

- Log status as `MISSED`.
- Schedule a **Retry Task** for +4 hours later.

5 Phase 5: Engagement AI Dashboard

5.1 Frontend Development (React)

- Build the "Flyout" style dashboard (referencing `Engagement-AI...png`).
- **Visualizations:**
 - Risk Score Trendline (using `Recharts/Chart.js`).
 - "Top 3 Risk Factors" highlighter.
 - Relationship Score gauge.